

Shakil Ahamed

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in LinkedIn 🐙 GitHub 📍 Mohakhali, Dhaka, Bangladesh

Professional Summary

Full-Stack Software Engineer specializing in **MERN Stack (Node.js, Express.js, React.js, Next.js)** with advanced expertise in **AI-driven system integration**. Experienced in designing **scalable backend architectures, RESTful APIs, authentication workflows, and database systems (PostgreSQL, MongoDB)**. Proven ability to build **production-ready SaaS platforms**, real-time systems, and **AI-enabled applications** with clean architecture and performance optimization.

Technical Skills

- **Backend Development:** Node.js, Express.js, Python, FastAPI, REST APIs, JWT Authentication, WebSockets
- **Frontend Development:** Next.js, React.js, HTML, CSS, Tailwind CSS
- **Databases:** PostgreSQL, MySQL, MongoDB, Firebase Firestore, SQLite, Schema Design, Query Optimization
- **Programming Languages:** Python, JavaScript(ES6+), SQL, C, TypeScript
- **Machine Learning & Deep Learning:** Scikit-learn, TensorFlow, Keras, XGBoost, LightGBM
- **Computer Vision:** OpenCV, YOLO, MediaPipe, SAM, U-Net, Stable Diffusion
- **LLM & Multi-Agent Integration:** OpenAI API, Hugging Face, LangChain, LlamaIndex, FastMCP
- **Cloud & DevOps:** Docker, Netlify, AWS, CI/CD Pipelines, Linux, Cloud Deployment
- **Engineering Practices:** Git, Agile Workflow, Testing, Jupyter, VS Code

Professional Experience

Full-Stack AI Engineer

BETOPIA GROUP (SOFTVENCE - SISTER CONCERN)

Dhaka, Bangladesh
January 2025 – Present

- Designed and implemented **scalable backend systems** using **FastAPI and Python**, developing secure **RESTful APIs** with structured architecture.
- Architected and optimized **PostgreSQL and MongoDB** database schemas, improving performance and ensuring data integrity.
- Developed responsive frontend interfaces using **React.js and Next.js** with performance-focused UI design.
- Integrated **LLM-based AI modules** and **Computer Vision pipelines** into production-grade web applications.
- Containerized applications using **Docker** and deployed solutions on **Render and Vercel** with CI/CD automation.
- Collaborated within cross-functional engineering teams following **Agile development** practices.

Selected Digital Systems & Platforms

- **Multi-Agent AI Platform:** Integrated image, voice, and text-based agents to support e-commerce assistance and digital content workflows. GitHub: [MultiAgent_with_MCP](#)
- **Mock Interview Feedback System (InPrep-AI):** AI-based assessment platform using computer vision and LLM feedback to support skill development and confidence evaluation. Live: [inprep.ai](#)
- **AI Student Portal:** Full-stack student management platform integrating AI features. Live: [zyura-e.com](#)
- **Medical Home Service Platform:** Full-stack service management system with authentication, scheduling, and cloud deployment. GitHub: [Medical_Home_Service](#)
- **Data Analytics Dashboards:** Business-focused Power BI analytics solutions translating raw data into KPIs and actionable insights.
- **AI Ad Generation System:** Shot-based AI ad video pipeline using GPT-4, MusicGen, and Stable Diffusion for end-to-end ad creation.
- **Car Body Wrap AI:** Computer vision pipeline for car segmentation (U-Net/SegFormer/SAM) combined with Stable Diffusion wrap generation.
- **Biovue:** AI-powered fitness recommendation system analyzing daily activities to provide personalized food and exercise suggestions.
- **Image Captioning Model:** Deep learning model generating human-like descriptions from visual input.

Core Competencies

- Digital Programme Design & Delivery
- Requirements Gathering & Functional Specification
- SDLC, Agile Execution & Milestone Planning
- Quality Assurance, Testing & UAT Coordination
- AI System Design & Deployment
- Computer Vision & Deep Learning
- Multi-Agent Orchestration & LLM Integration
- Data Pipeline & MLOps Practices
- Cross-functional & Stakeholder Coordination
- Technical Documentation & Reporting

Education

Gopalganj Science and Technology University
B.Sc. in Computer Science and Engineering

Bangladesh
2019 – 2022

Certifications

- Data Analysis with Python — EDGE, BSMRSTU (2024)
- Mobile & Game Application Development — ICT Division, Bangladesh (2022)